The Vocabulary of Sound: A Codebook Analysis

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Abstract

Neural audio codecs, such as Encodec, serve as the foundational building blocks for modern generative audio by compressing continuous waveforms into discrete latent codes. However, the internal logic by which these models allocate their finite vocabulary (codebook) to different acoustic distributions remains a black box. This study investigates codebook utilization across a diverse 13-class dataset, ranging from complex symphonic music to generated simple sound files (Sine, Square, Sawtooth, and Half-rectified waves).By quantifying the latent space through Shannon Entropy, the Gini Coefficient of token distribution, and the Temporal Code Change Rate (TCCR), we reveal distinct "Acoustic Regimes" in the AI's internal representation. Our findings demonstrate a latent collapse in steady-state and deterministic signals, where the model utilizes a minimal, repetitive subset of tokens despite the availability of 1024 codes. Conversely, structured audio like music exhibits high vocabulary inequality , relying on specialized anchor tokens to maintain most of the information.Also we found that the Shannon Entropy of codes is highly related to the TCCR. These results prove that Encodec's compression efficiency is limited and is driven by temporal predictability , offering a strong empirical foundation for designing future adaptive, content-aware codebook architectures in generative audio systems. [1]

1 Background and Literature Review

1.1 Neural Audio Codec

Neural audio codecs aim to compress continuous audio waveforms into low-bitrate discrete representations via end-to-end learning while maintaining high-fidelity reconstruction quality. Unlike traditional audio coding standards (such as MP3, AAC, and Opus) that rely on hand-crafted signal processing modules and psychoacoustic models, neural codecs automatically learn optimal compression strategies in a data-driven manner.

[1] All the code and data is available at `https://github.com/Wu0219/GenAI_Final` , audio is submitted together with the paper and in the google drive.

This paradigm shift began with the Vector Quantized Variational Autoencoder (VQ-VAE) proposed by van den Oord et al. [1], which addressed the bottleneck of converting continuous spaces to discrete ones through discrete latent representation learning. Subsequently, Google's SoundStream [2] further applied this framework to the audio domain and introduced the Residual Vector Quantization (RVQ) mechanism for the first time, laying the theoretical foundation for modern neural codecs to surpass traditional algorithms at extremely low bitrates.

1.2 EnCodec and Its Evolution

EnCodec [3], a representative model proposed by Meta AI in 2022, aims to balance high fidelity, real-time encoding, and ultra-low latency. Its core architecture consists of three integrated components: a convolutional encoder based on the SEANet structure, a residual vector quantizer (RVQ) with progressive refinement, and a mirror-symmetric decoder. EnCodec supports a configurable temporal resolution of 75 Hz, enabling high-quality reconstruction across an extremely wide bandwidth range.

Following the release of EnCodec, stronger SOTA models such as DAC (Descript Audio Codec) [4] have emerged in the audio codec field. By improving the training strategy of RVQ-GAN, DAC further enhances codebook utilization and perceptual quality. However, although these models perform excellently on reconstruction metrics, the intrinsic dynamics of their underlying discrete dictionaries when handling varying acoustic complexities remain to be thoroughly explored.

1.3 Codebook Collapse and Non-stationarity

Despite the remarkable success of neural decoders, the vector quantization process is often plagued by codebook collapse. This means that a large portion of vectors in the codebook (dead codes) are never selected or updated during training, severely limiting the model's representational capacity.

Although prior studies such as VQGAN [5] have attempted to mitigate this issue by introducing heuristic strategies like commitment loss and codebook reset, they lack a solid theoretical foundation. Recently, Yang et al. [6] proposed a breakthrough perspective, identifying the root cause of codebook collapse as the non-stationarity of encoder updates. As encoder parameters drift during training, the latent space distribution shifts, causing unused codebook vectors to gradually deviate from the data distribution and ultimately become inactive.

2 Justification of the Study

This study investigates codebook utilization in the discrete latent space of neural audio codecs. Its justification lies primarily in the following three dimensions:

2.1 Filling the Academic Gap in Black Box Representations

Although models like EnCodec perform excellently on macro-level reconstruction metrics such as Signal-to-Noise Ratio, SNR, or perceptual quality ViSQOL, the internal logic of their codebook allocation remains largely a black box Most current research focuses on improving reconstruction fidelity, yet few studies deeply analyze how models allocate their limited discrete resources across acoustic signals of varying complexity, from pure mathematical sine waves to highly random natural noise. This study introduces the Gini Coefficient and Temporal Code Change Rate (TCCR) to quantify the wealth gap and temporal volatility of codebook usage, offering a novel experimental perspective for understanding the inductive bias of generative audio models.

2.2 Enabling Efficient Generative Architectures

The findings of this study offer significant guidance for designing next-generation audio generation models. Currently, most neural codecs employ fixed bit-rate allocation schemes (e.g., 8-layer RVQ with uniform quantization), which are computationally inefficient. By demonstrating that physically high-entropy signals can exhibit logical collapse in latent space (such as the low entropy and high inequality observed in natural noise), this work lays the foundation for developing semantic-aware dynamic codebook allocation algorithms. Such techniques will enable future models to automatically reduce codebook depth for simple signals while fully leveraging capacity for critical, complex signals (e.g., vocals or symphonies), thereby greatly optimizing inference efficiency and storage costs for generative AI without compromising perceptual quality.

3 Methodology

3.1 Dataset

To comprehensively evaluate the performance of EnCodec codebooks across different acoustic modalities, this study constructed a comprehensive dataset comprising 13 categories of audio samples. These samples are primarily derived from three sources: online acquisition via the Freesound API, from classical music recordings, and algorithmic synthesis using Python scripts. Detailed categorization, sources, and acquisition parameters are presented in Table 1.

Table 1: Dataset Classification, Sources, and Parameter Specifications

ID	Category Name	Acquisition/Synthesis Method	Search Keywords / Algorithm Parameters
Natural & Complex Acoustic Scenes			
01	Synthetic	Freesound API	"8-bit"
02	Speech	Freesound API	"speech"
03	Instrument	Freesound API	"piano"
04	Percussion	Freesound API	"drum"
05	Crowd	Freesound API	"crowd talking"
06	Nature	Freesound API	"rain"
07	ComplexMusic	Beethoven classical recordings	*Symphony No. 9*
09	Impulsive	Freesound API	"Gun"
Algorithmic Synthetic Random Noise (Category 08)			
08a	White Gaussian	NumPy Random	Gaussian white noise ($\mu = 0, \sigma = 0.5$)
08b	Uniform White	NumPy Random	Uniform white noise ($[-0.5, 0.5]$)
08c	Brownian	Cumsum Random	Brownian noise (red noise), simulating $1/f^2$ spectrum
Algorithmic Synthetic Deterministic Waveforms			
10	Sine	NumPy Signal	Pure sine wave, $f \in [220, 880]$ Hz
11	Square	SciPy Signal	Square wave (containing odd harmonics), $f \in [220, 880]$ Hz
12	Sawtooth	SciPy Signal	Sawtooth wave (containing all harmonics), $f \in [220, 880]$ Hz
13	HalfSine	NumPy Non-linear	Half-wave rectified sine wave, $f(t) = \max(0, \sin(\omega t))$

To ensure the representativeness and statistical significance of the experimental results, we conducted a comprehensive audit of the final dataset. Table 2 details the sample distribution, total duration, and average sample length across 13 acoustic modalities.

Table 2: Summary of Dataset Sample Distribution and Duration Statistics

ID	Acoustic Category	Samples (N)	Total Duration (s)	Avg. Duration (s)
Natural & Real-world Audio				
01	Synthetic	30	237.62	7.92
02	Speech	30	242.77	8.09
03	Instrument	30	222.22	7.41
04	Percussion	30	182.05	6.07
05	Crowd	12	126.91	10.58
06	Nature	30	355.55	11.85
07	ComplexMusic	399	3990.02	10.00
08	Noise (Algorithmic)	45	450.00	10.00
09	Impulsive	30	232.93	7.76
Deterministic Waveforms				
10	Sine	15	150.00	10.00
11	Square	15	150.00	10.00
12	Sawtooth	15	150.00	10.00
13	HalfSine	15	150.00	10.00
Total	**13 Categories**	**696**	**6640.07**	**9.54**

Note: The dataset comprises 696 samples with a cumulative audio duration of approximately 1.84 hours. The ComplexMusic category was obtained via large-scale slicing to ensure robustness in complex audio feature analysis.

The full dataset is available at Google Drive, and the first audio of each category is submitted together with the paper.

3.2 Code Extraction

Before code extraction, all input audio (covering both `.wav` and `.ogg` formats) is standardized using the `librosa` library. Audio is uniformly resampled to $24,000$ Hz and converted to mono to ensure statistical consistency of the input tensors. Subsequently, we normalizes the amplitude to meet the numerical range requirements of the Encodec encoder.

The preprocessed audio tensor is fed into the EnCodec encoder network. In this experiment, we strictly limit the target bandwidth to 6.0 kbps. This configuration is critical as it determines the depth of the RVQ layers. At 6.0 kbps, the model describes the signal using the first 8 quantizers, generating a feature matrix of dimensions $[8, T]$, where T is the number of time frames.The extracted feature matrix is serialized into a `.npy` file using NumPy.

3.3 Shannon Entropy

This experiment uses Shannon entropy to describe the richness of the code, with the specific formula as follows, where K is the total capacity of the dictionary. Here, $K = 1024$. $p(x_i)$ is

the probability that the i-th code appears in the current segment.

$$H(X) = -\sum_{i=1}^{K} p(x_i) \log_2 p(x_i)$$

3.4 Gini Coefficient

To quantify the concentration and inequality in the distribution of codes within the feature space, we introduce the Gini Coefficient. Traditional Gini coefficients suffer from severe statistical bias when dealing with extremely sparse codebooks. Therefore, we first filter out inactive codewords with zero frequency and compute the inequality index solely over the set of activated codewords. Let $X = \{x_1, x_2, \ldots, x_n\}$ denote the frequencies of activated codewords (i.e., those with frequency greater than zero), sorted in ascending order ($x_1 \leq x_2 \leq \cdots \leq x_n$), where n is the total number of distinct activated codewords. The Robust Gini Coefficient G_{robust} is calculated as follows:

$$G_{\text{robust}} = \frac{\sum_{i=1}^{n} (2i - n - 1) x_i}{n \sum_{i=1}^{n} x_i}$$

where i represents the rank index after sorting the codeword frequencies in ascending order. When G_{robust} approaches 0, it indicates that activated codewords are used extremely uniformly; conversely, when it approaches 1, it signifies that a very small minority of codewords account for the vast majority of occurrences.

3.5 Temporal Code Change Rate

To measure the dynamic fluctuation of acoustic feature sequences along the time dimension, we define the Temporal Code Change Rate (TCCR). This metric quantifies the local temporal instability of an audio sequence by calculating the transition frequency of feature states between adjacent time steps.

Given a one-dimensional feature sequence $X = \{x_1, x_2, \ldots, x_T\}$ of length T, where x_t denotes the codeword index at the t-th time frame, TCCR is defined as:

$$\text{TCCR} = \frac{\sum_{t=2}^{T} \mathbb{I}(x_t \neq x_{t-1})}{T - 1},$$

where $\mathbb{I}(\cdot)$ is the indicator function that equals 1 when a codeword transition occurs between adjacent frames (i.e., $x_t \neq x_{t-1}$), and 0 otherwise. The denominator $T - 1$ normalizes the count by the maximum possible number of transitions in the sequence. TCCR ranges from 0 to 1, with higher values indicating more intense temporal variations at the feature level.

4 Results and Discussions

4.1 Codebook Sparsity & Global Utilization

This section focuses on the utilization efficiency and global distribution patterns of EnCodec's RVQ codebooks. Combined with chart analysis, we find that the mapping of discrete acoustic features in the latent space exhibits high imbalance.

As shown in Figure 1, although the theoretical maximum codeword capacity of a single quantization layer is 1024, the number of active codes for most audio categories during actual encoding is far below this theoretical limit. In particular, for simple acoustic signals containing a single harmonic or periodic structure (such as sine and square waves), their representation often relies on only a very small number of discrete codewords. This phenomenon indicates that fundamental acoustic features exhibit strong sparsity in the RVQ discrete latent space.

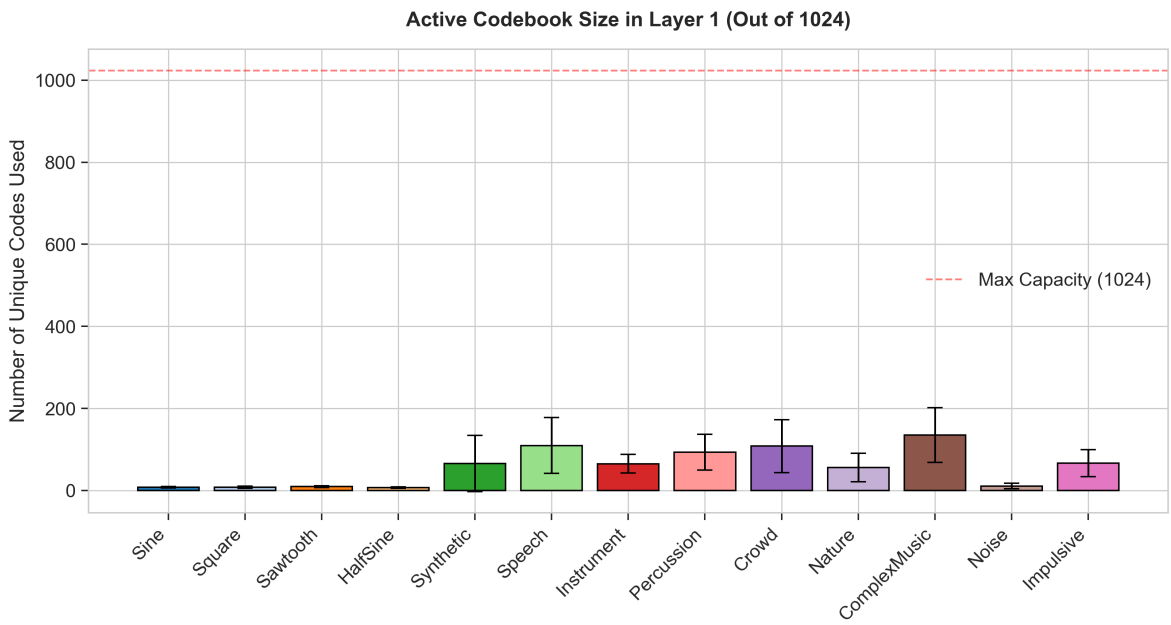

Figure 1: Active Codes

Global frequency distributions and rank-frequency plots further reveal overall statistical patterns in codebook utilization as shown in Figure 2. Even with highly extreme synthetic audio, the usage frequencies of global codewords exhibit typical imbalance. The data shows pronounced head concentration: a tiny fraction of core codewords carries the vast majority of acoustic mappings, accompanied by a significant long-tail effect, with absolute zero-frequency dead codes clearly observed at the tail. This indicates that existing codebook capacity entails a degree of parameter redundancy from a global perspective.

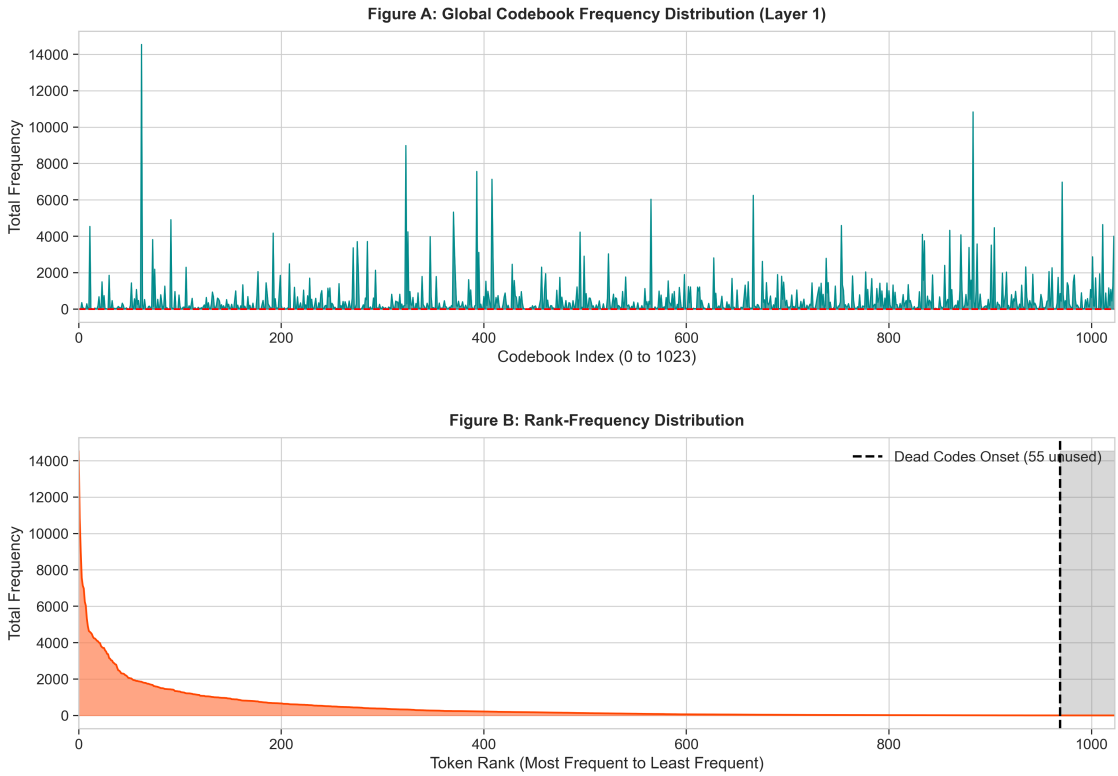

Figure 2: Global Codebook utilization

As shown in the distribution in Figure 3, the fundamental usage patterns of codes exhibit significant cross-domain variations across different audio categories. Complex audio (e.g., Crowd and ComplexMusic) activates a broader range of codewords, indicating the need for richer discrete representations to reconstruct intricate acoustic textures and wide dynamic ranges. Nevertheless, even within these high-dimensional categories, local densities still display pronounced alternating peaks and valleys, further confirming that non-uniform local clustering is a pervasive characteristic of the RVQ discrete encoding process.

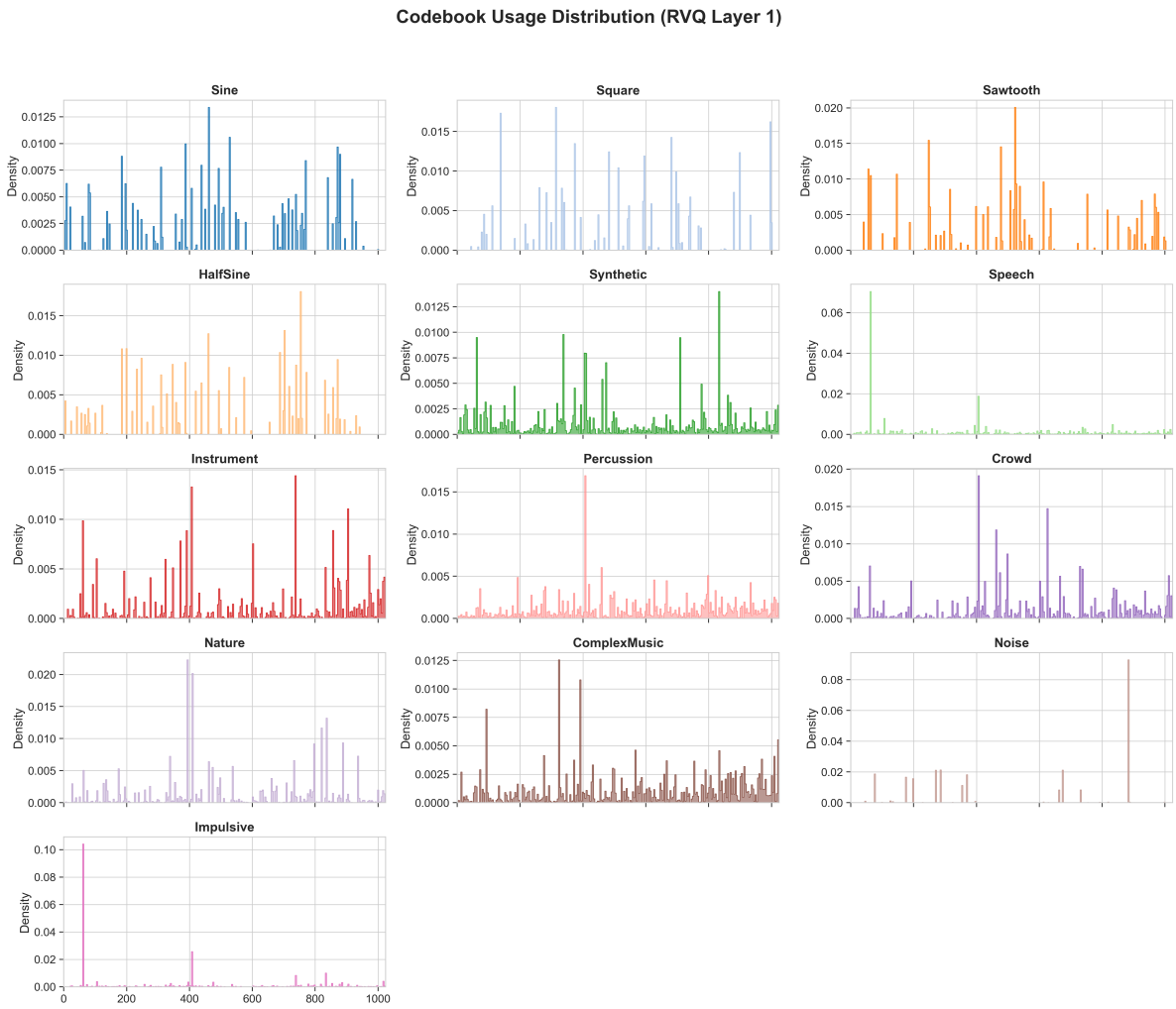

Figure 3: Full Distributions in RVQ layer 1

4.2 Spatial Complexity & Information Capacity Across RVQ Layers

As shown in the entropy trajectory line chart in Figure 4 and the global heatmap in Figure 5, the average Shannon entropy across all audio categories exhibits a significant increasing trend with greater RVQ hierarchical depth.

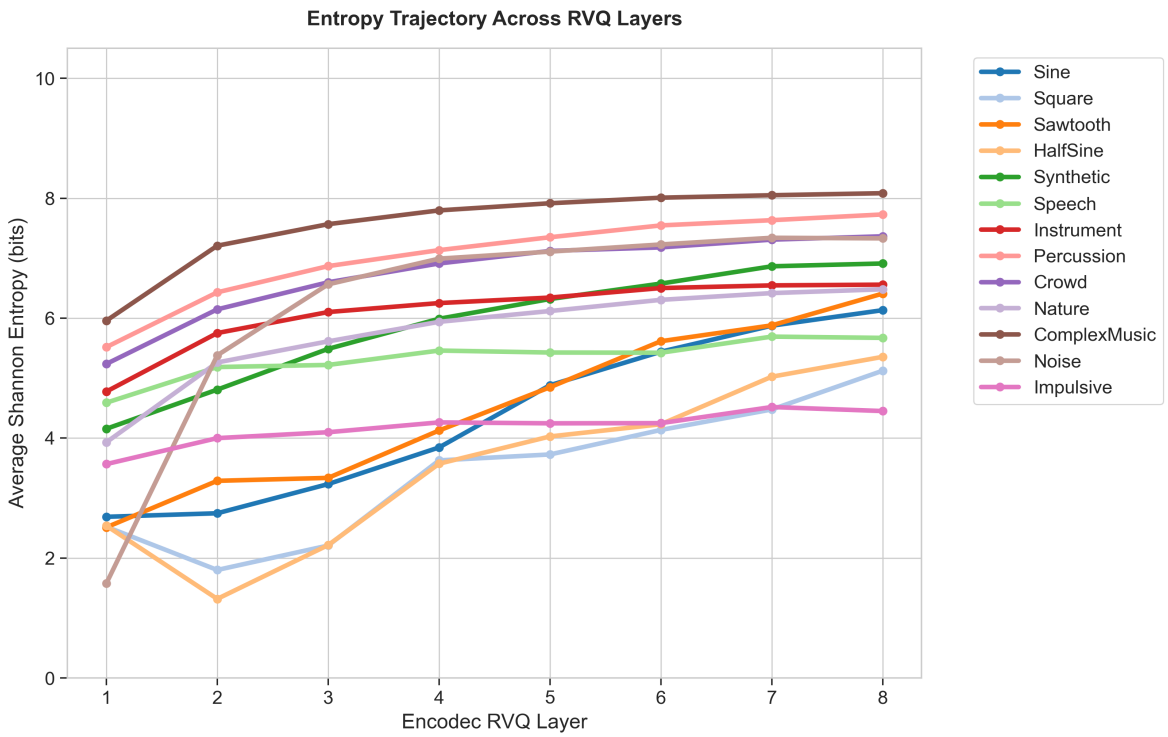

Figure 4: Entropy Trajectory

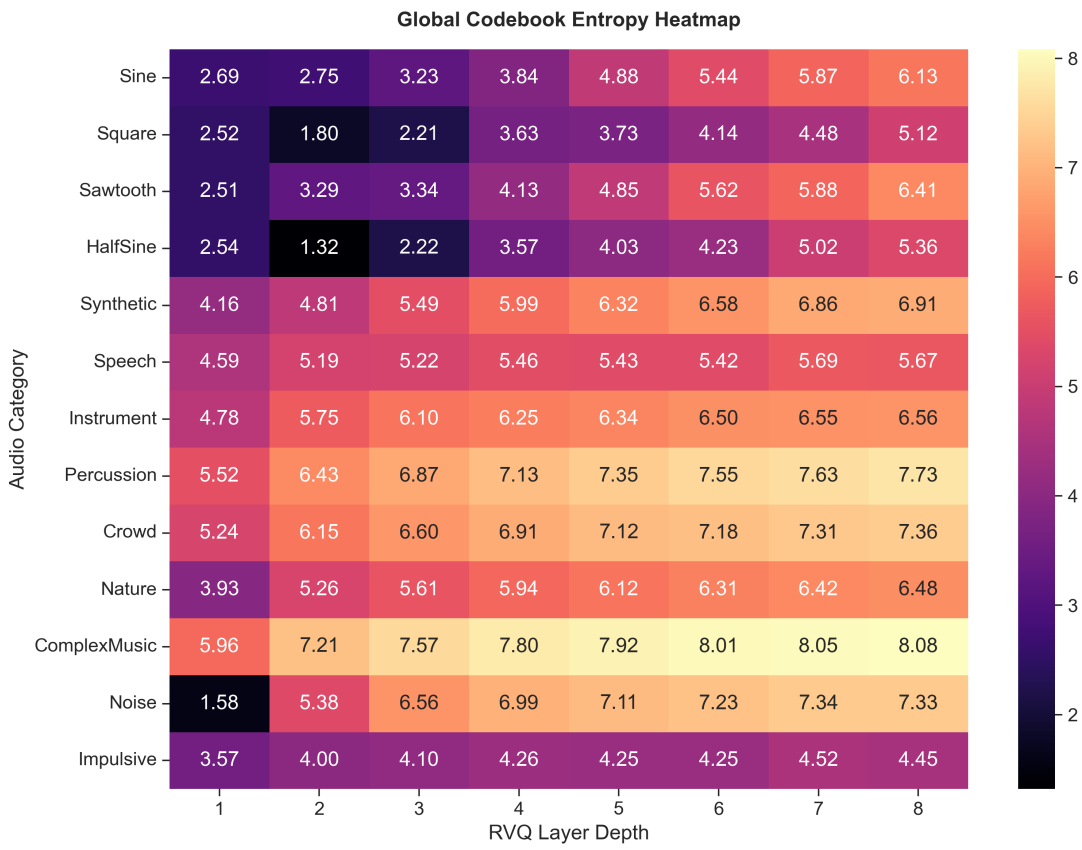

Figure 5: Entropy Heatmap

Within the same quantization level, different acoustic signals exhibit markedly distinct information capacities, forming a clearly stratified hierarchy. Natural sounds and highly mixed complex audio (such as ComplexMusic and Percussion) display extremely high spatial complexity, with deep-layer entropy values significantly exceeding those of other categories and approaching the theoretical upper bound of codebook distribution (10bits). In contrast, audio featuring deterministic periodicity or simple synthesized waveforms (such as Sine and Square waves) consistently resides at the bottom tier of information capacity.

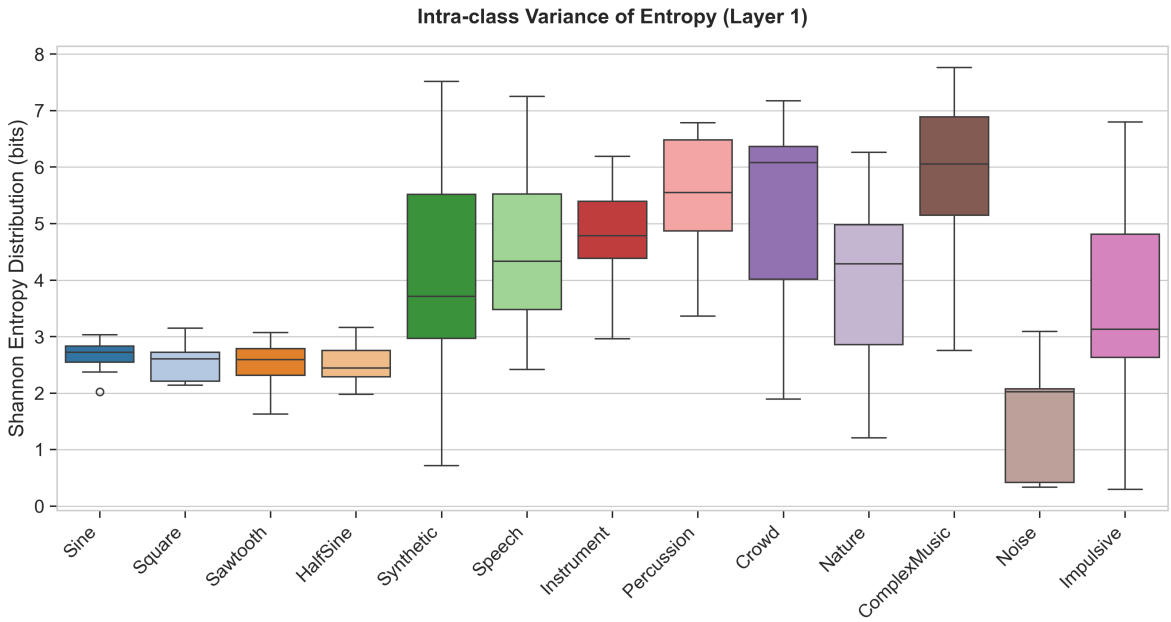

Figure 6: Variance Boxplot

Beyond inter-class differences, the box plots in Figure 6 further reveal Shannon entropy fluctuations within individual audio categories. The study finds that signals with highly stationary stochastic characteristics (e.g., Noise) or strict physical determinism (e.g., Sine) exhibit minimal intra-class variance, indicating highly consistent latent-space mapping patterns across their internal samples. In contrast, synthetic audio and human speech display substantial spans in entropy values and high data dispersion.

4.3 Vocabulary Inequality

As shown in Figure 7, complex audio exhibits a combination of high entropy and relatively low-to-mid Gini coefficients(But still very high). This indicates that complex acoustic textures achieve efficient representation by activating a large subset of codewords while maintaining relatively balanced usage frequencies, rather than relying on the stacking of single features.

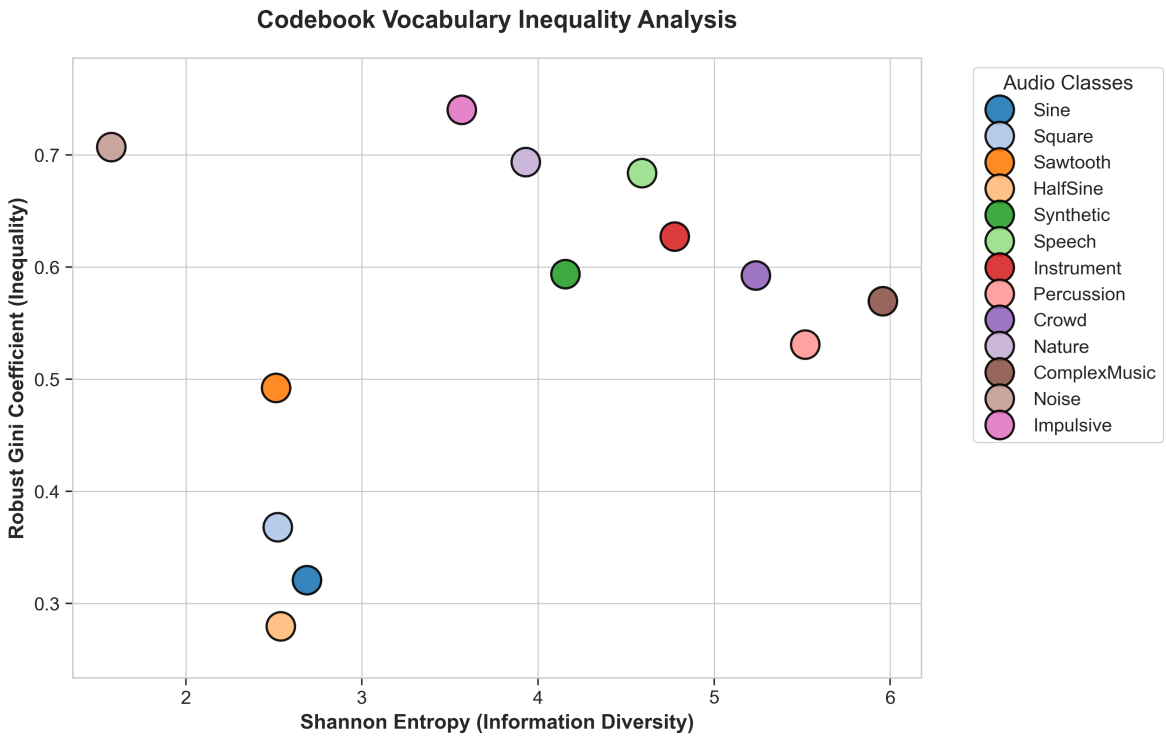

Figure 7: Gini Analysis

4.4 Code Dynamics Over Time

Figure 8 shows clear clustering in the (TCCR, H) space across audio categories. ComplexMusic lies in the upper-right region, combining high entropy and high volatility, and thus marking the highest representational pressure. By contrast, periodic signals (Sine, Square, Sawtooth) show low entropy (≈ 2.5 bits) but very high fluctuation (TCCR > 0.9), indicating compact vocabularies with rapid temporal switching. Noise and Impulsive sounds cluster in the lower-left region, suggesting that Layer-1 RVQ relies on relatively stable codeword patterns for broadband statistics, while finer randomness is deferred to residual layers.

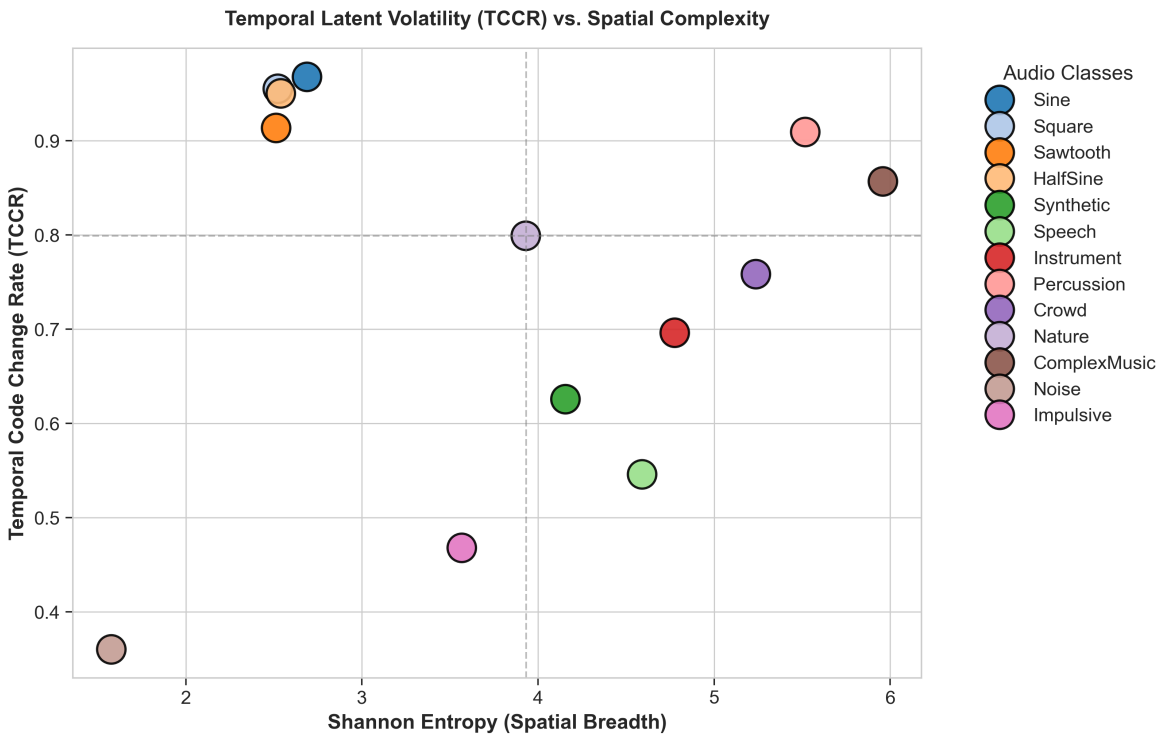

Figure 8: Audio TCCR vs Entropy

Experiments revealed a significant positive correlation between spatial complexity and temporal volatility (Pearson $r = 0.82, p < 0.001$). This demonstrates that the extent of latent space resource utilization is highly consistent with the frequency of state transitions along the time axis in physical terms.

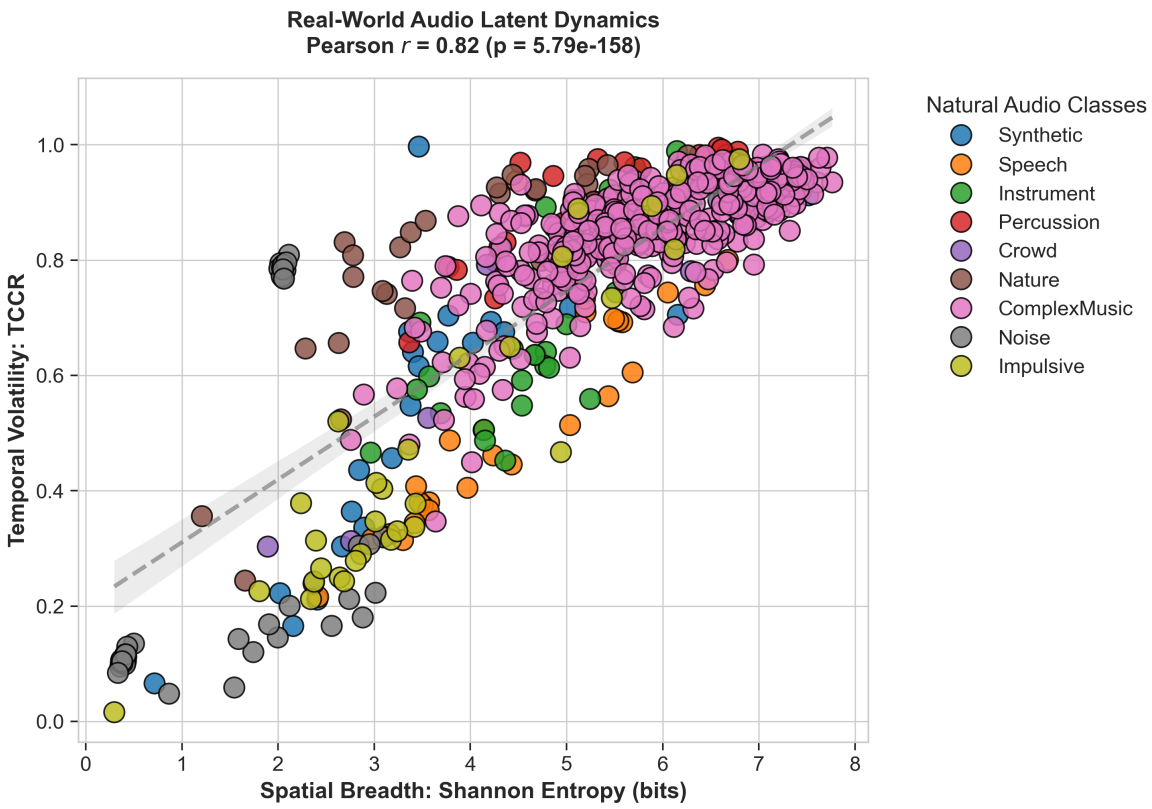

Figure 9: Natural Audio TCCR vs Entropy

5 Conclusion

This study provides a systematic analysis of codebook utilization in the discrete latent space of EnCodec. By jointly leveraging Shannon entropy, the Robust Gini coefficient, and the proposed Temporal Code Change Rate (TCCR), we uncover several fundamental structural properties of neural audio representations.

First, codebook usage is highly imbalanced: despite a large theoretical capacity limit, only a small subset of codebooks is actively and frequently used, forming a head–tail distribution. This indicates substantial redundancy and a strong inductive bias in the learned discrete space. Second, distinct acoustic regimes emerge across different signal types. Complex audio (e.g., music and crowd sounds) exhibits high entropy and broad codeword activation, whereas deterministic or periodic signals (e.g., sine and square waves) show clear latent collapse, relying on only a minimal subset of tokens.

Importantly, we observe a strong positive correlation between entropy and TCCR (r=0.82), suggesting that spatial information complexity is tightly coupled with temporal variability. This MIGHT reveals a key underlying mechanism: codebook utilization is primarily driven by temporal non-stationarity rather than raw spectral complexity.

Overall, our findings suggest that current neural audio codecs implement a temporally driven, highly non-uniform discrete representation. This provides empirical support for future designs of content-aware, adaptive codebook allocation strategies, which could significantly

improve both efficiency and representational capacity in generative audio systems.

6 Limitation

Due to constraints in capability and time, this study still has several limitations. First, the data selection may not cover all necessary ranges, potentially leading to certain biases in the results. Second, the calculation and validation of TCCR rely heavily on codebook-based TCCR, which may cause its distribution and variations to be inherently linked to codebook entropy. Furthermore, even if a correlation is established, it does not imply causality; therefore, further experiments are needed to investigate the specific relationship and underlying causes.

7 Future Directions

Future work should focus on building larger and higher-quality datasets with finer-grained category labels, validating the robustness of TCCR through additional metrics and audio-level ablation studies rather than relying only on codebook signals, developing semantic-aware dynamic RVQ allocation to improve compression efficiency without sacrificing perceptual quality, and extending the analysis to a broader set of neural audio codec models.

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